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Bihar Engineering University, Patna

B.Tech 4th Semester Exam., 2022

Time: 03 Hours

Code: 101406 ***Subject: Materials, Testing and Evaluation*** ***Full Marks: 70***

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory.*
(v) *Assume any data not given.*

Q.1 Choose the correct answer of the following (any seven) : **[14]**

(a) Brittle fracture is more dangerous than ductile fracture because
(i) of no warning sign
(ii) crack propagates at very high speed
(iii) no need for extra stress during crack propagation
(iv) All of the above

(b) In Vicat's apparatus, the cement paste is said to be of normal consistency, if the rod penetrates by
(i) 33 mm to 35 mm
(ii) 23 mm to 25 mm
(iii) 5 mm to 10 mm
(iv) 3 mm

(c) Gap grading is one
(i) in which the particles fall within a narrow limit of size fractions
(ii) in which one or more intermediate fractions are absent
(iii) which combines different fractions of fine and coarse aggregates
(iv) in which all the particles are of uniform size

(d) The fineness modulus of an aggregate sample is 5.0, its representative geometric mean size is
(i) 10.00 mm
(ii) 4.75 mm
(iii) 2.36 mm
(iv) 1.18 mm

(e) For ordinary Portland cement, the maximum expansion by Le Chatelier's method should not exceed
(i) 10 mm
(ii) 7.5 mm
(iii) 5 mm
(iv) 2 mm

(f) A compacting factor of 0.88 or a fresh concrete sample indicates a mix of
(i) high workability
(ii) medium workability
(iii) low workability
(iv) very low workability

(g) The angle of internal friction, is least for
(i) angular-grained loose sand
(ii) angular-grained dense sand
(iii) round-grained loose sand
(iv) clays

(h) The resistance to flow of bitumen is measured by
(i) Flash and fire test
(ii) penetration test
(iii) viscosity test
(iv) ductility test

(i) The shear strength of a soil
(i) is directly proportional to the angle of internal friction of the soil
(ii) is inversely proportional to the angle of internal friction of the soil
(iii) decreases with increase in normal stress
(iv) decreases with decrease in normal stress

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Q.2 Solve both questions : **[14]**

(a) What is proof stress? What is the difference of behaviour of a mild steel bar, HYSD steel bar and TMT steel bar? What is called necking?

(b) How the yield strength and ultimate tensile strength of a steel specimen is determined from laboratory test? What is gauge length? What do you mean by percentage elongation and what is the limit as per IS code?

Q.3 Solve both questions : **[14]**

(a) Table 1 shows the details of weights of sample retained on various sieves. The total weight of the coarse aggregate is 10 kg and the fine aggregate is 1 kg. Determine the fineness modulus of the coarse and fine aggregate.

Sieve	Coarse aggregate Weight retained (in kg)	Fine aggregate Weight retained (in gm)
80 mm	0	
40 mm	0	
20 mm	4	
10 mm	3	0
4.75 mm	3	30
2.36 mm	0	135
1.18 mm		135
600 µm		385
300 µm		225
150 µm		90

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(b) Define maturity of concrete. How the compressive strength of concrete varies with maturity? Explain segregation, bleeding and laitance in green concrete. Explain elongation index and flakiness index of aggregates.

Q.4 Solve both questions : **[14]**

(a) Explain various factors that affect workability of concrete. State various tests along with their suitability for characterizing the specific aspects of workability.

(b) Explain chemical composition of ordinary Portland cement. Explain the test procedure of determining the specific gravity and soundness of cement. Explain the factors on which the soundness depends.

Q.5 Solve both questions : **[14]**

(a) Explain the procedure to determine the specific gravity and moisture content of sand. Define crushing value, impact value and ten% fines value of aggregate. What is the importance of ten% fines value of aggregate?

(b) What are the quality requirements of water for mixing and curing of concrete? Is it possible to use recycled water from industry or sea water for curing? Explain various measures to protect corrosion of reinforcement in concrete.

Q.6 Solve both questions : **[14]**

(a) What are the causes of unsoundness in cement? Explain the tests commonly used for assessing the soundness of cement.

(b) Why is compaction so important in concrete? Explain common means of compaction.

(c) Explain curing of concrete. Elaborate various methods of curing of concrete.

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Q.7 Solve all three questions : **[14]**

(a) What are the physical properties of soil? Classify different types of soil based on its plasticity index. State the laboratory procedure for determination of liquid limit of soil.

(b) What is Mohr's circle? What is the theory of failure as per Mohr's circle? Explain the direct shear test of soil. What is the need of triaxial shear test of soil?

Q.8 Solve both questions : **[14]**

(a) What causes fatigue failure of material? What is the difference between creep and fatigue failure? What is stress ratio in fatigue? What is mean stress correction? What is Soderberg line?

(b) How can the ductile fracture be identified? Why brittle failure is dangerous? What causes transgranular facture? How can the toughness be determined from test? What is stress intensity factor in fracture mechanics?

Q.9 Solve both questions : **[14]**

(a) What is the meaning of 60/70 bitumen? What is the softening point of bitumen? Explain the ductility test of bitumen.

(b) Explain roller compacted concrete and ready mixed concrete.

(c) What do you mean by target strength? Explain the procedure of M25 grade concrete mix design.

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